

Summary

- **Machine Learning Researcher** specializing in **image processing**, having established a **pre-trained GAN for limited-labeled brain MRI segmentation**, advancing **medical AI research**.
- Strong foundation in **machine learning theory and practice**, with experience in **model development, training, and optimization**, and a clear focus on contributing to **cutting-edge AI projects** and **real-world applications**.
- **Active open-source contributor** in AI, collaborating on **public and volunteer projects**, reflecting commitment to **innovation, collaboration, and advancement of the AI community**.

Education

Bachelor of Computer Engineering - GPA: 17.88/20 (equivalent to 3.57/4.00)
Zand Institute of Higher Education

Sep 2021 - Aug 2025
Shiraz, Iran

Visiting Student
University of Science and Culture

Feb 2023 - Jul 2024
Tehran, Iran

Research Interests

- Machine learning
- Computer Vision
- Deep Learning
- Computational Biology
- Natural Language Processing
- Scientific Computing

Publication

Pre-Trained Generative Adversarial Network for Limited-Labeled Brain MRI Segmentation engrXiv, DOI: [10.31224/6504](https://doi.org/10.31224/6504).
Mehdi Karami[✉], Betsabeh Tanoori

Open Source Projects

- **Retrieval-Augmented Generation (RAG) for Small Language Models (SLMs)**
 - Implements a RAG pipeline which including text chunking, embedding, and vector retrieval to enable efficient generation and inference with local Small Language Models (SLMs) as well as standard language models.
 - Stack: Pytorch, FAISS, Transformers (HuggingFace), Sentence-Transformers, LangChain, PyMuPDF4LLM, ...
- **AI Sight Quest (Bachelor's Project)**
 - Utilizes the BERT language model to extract text from images and answer questions about documents.
 - Stack: BERT-SQuAD, Swift, SwiftUI, SwiftData, Apple's Vision and Speech frameworks, TipKit, Protocol-Oriented Programming (POP), Model-view-viewmodel (MVVM), ...
- **Cyberattack Detection and Anomalous Behavior Analysis**
 - Utilizes an LSTM-based RNN to detect cyberattacks from anomalous network traffic, including DDoS, port scanning, and brute-force attempts.
 - Stack: TensorFlow/Keras, Scikit-learn, PCA, ROC Curve, ...
- **Solar Power Generation Prediction**
 - Implements ML models (Linear, Decision Tree, Random Forest, Gradient Boosting, MLP, DNN) to predict solar power generation, evaluated via MAE, MSE, R^2 , and confusion matrix.
 - Stack: TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn, ...
- **NexumAI**
 - Provides an interface for training models (e.g., image classification, segmentation) without technical expertise, letting users obtain trained models by supplying datasets and configurations.
 - Stack: TensorFlow, PyTorch, Torchvision, PyQt6, ...

Additional projects can be found at: [GitHub account](#).

Work Experience

iOS Developer (Full-time, On-site)
Round Table Apps (Sydney, Australia)

Jun 2023 – Jun 2024
Development Office, Tehran, Iran

- Engineered **Blossom – Save and Invest**, a consumer-facing financial application for the **Australian** and **New Zealand** markets, serving **25K+** active users.
- Maintained **MailPlus – MP Operator**, a post-transportation application for **MailPlus Australia**, supporting **300+** owner-operators and **30K+** customers.
- Developed and refactored **Resilience Box**, an **international** wellbeing and resilience platform.

Technical Contributions:

- Researched the development of **AR** and **VR** technologies in **healthcare** using **visionOS**.
- Conducted in-depth analysis of **data flows** to identify and model **user behavioral patterns** across various scenarios.
- Applied key design patterns such as **MVC**, **MVVM**, **VIP**, and **Repository**.
- Boosted development efficiency by up to **40%** through the creation of custom file templates.

Teaching Experience

Teaching Assistant (TA)

Zand Institute of Higher Education, Shiraz, Iran

- Fall 2024:**
 - Human-Computer Interaction
 - Digital Electronics
- Spring 2025:**
 - Fundamentals of Computer and Programming
 - Digital Electronics
 - Computer Architecture

Technical Skills

- Programming & Scripting:** Python, C/C++, Julia, MATLAB, Swift, Objective-C, SQL, Bash, LaTeX, ...
- Deep Learning:** Neural Networks, Image Processing, Natural Language Processing, Transformers, GAN, RAG
- Tools & Platforms:** Jupyter, Google Colab, GitHub, GitLab, Command Line, Linux, macOS, Jira, Confluence, Trello
- Data Science & ML:** TensorFlow/Keras, PyTorch, Scikit-learn, OpenCV, HuggingFace, NumPy, Pandas, ...
- Software Development:** Git, CI/CD, Design Patterns, SOLID, iOS Development, Unit Testing, TDD, BDD, Agile
- Databases & Cloud:** MLOps, Vector DBs, Relational DBs (SQLite, PostgreSQL), Docker, Kubernetes, Query Optimization

Volunteering

Workshop on Git and GitHub

Zand Institute of Higher Education – [Certificate](#)

Dec 2024

K. N. Toosi University of Technology – [Certificate](#)

May 2024

Honors & Awards

Winning First Place in the C++ Programming Language Competition
Zand Institute of Higher Education

May 2022
Shiraz, Iran

Hobbies

- Swimming
- Traveling
- Cooking International Cuisine
- Esports Gaming (LOL)